TANVIR AHMED

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RESEARCH INTERESTS

Privacy-Preserving Wireless Sensing, Signal Processing, Applied Cryptography.

EDUCATION

Cornell University — Ithaca, NY 14850, US
Ph.D. in Information Science (GPA: **4.18**/4.3) — Aug 2023 - Aug 2028 *(Expected)*

- Research: Designing novel and secure wireless sensing systems for next-generation (6G) networks with applied cryptography and normative privacy frameworks. Relevant publications: P3, P2, C4. Collaborators: Thijs Roumen, Rafael Pass, Helen Nissenbaum.
- Committee: Rajalakshmi Nandakumar (Chairperson), Deborah Estrin, and Noah Snavely.
- Digital Life Initiative Doctoral Fellow and PiTech Impact Fellow at Cornell Tech.

Bangladesh University of Engineering and Technology (BUET) — Dhaka, Bangladesh
M.Sc. in Electrical & Electronic Engineering (GPA: **3.83**/4.00) — Apr 2019 - Jul 2023

- Thesis: Image Super-Resolution Using Wavelet Residual Convolutional Neural Networks [open access online]. Committee: S. M. Mahbubur Rahman (Chairperson), Shaikh Anowarul Fattah, Mohammed Imamul Hassan Bhuiyan, Md. Hasanul Kabir, and Md. Aynal Haque.

Bangladesh University of Engineering and Technology (BUET) — Dhaka, Bangladesh
B.Sc. in Electrical & Electronic Engineering (GPA: **3.82**/4.00) — Feb 2015 - Apr 2019

- Thesis: Traffic Sign Detection in Streaming Videos. Advisor: S. M. Mahbubur Rahman.

PUBLICATIONS†

[P3] mmFHE: mmWave Sensing with End-to-End Fully Homomorphic Encryption
Ahmed, T., Gao, Y., Armouti, A., & Nandakumar, R. — [arXiv:2603.22437]

- In Proc. of the *32nd Annual International Conference on Mobile Computing and Networking*, October 26-30, 2026, Austin, Texas, USA. (ACM MobiCom'26)
- Acceptance rate: 14.3% (89 out of 619 papers accepted).

[P2] PrivyWave: Privacy-Aware Wireless Sensing of Heartbeat
*Gao, Y., *Ahmed, T., Chang, Z., Roumen, T. & Nandakumar, R. — [arXiv:2511.02993]

- Under review.

[P1] SoilSound: Smartphone-based Soil Moisture Estimation
Gao, Y., Ahmed, T., He, S., Cheng, Z. & Nandakumar, R. — [arXiv:2509.09823]

- Under review.

[C4] VitalHide: Enabling Privacy-Aware Wireless Sensing of Vital Signs
*Gao, Y., *Ahmed, T., Chang, Z., Roumen, T. & Nandakumar, R. — [open access online]

- In Proc. of the *26th International Workshop on Mobile Computing Systems and Applications*, February 26-27, 2025, California, USA. (ACM HotMobile'25)

*equal contribution.

†Google Scholar: https://scholar.google.com/citations?user=YVYJl3QAAAAJ&hl=en

- Acceptance rate: 40%.
- News: Poster @ NYC Privacy Day at Google – Fall 2024, Cornell Tech News, Cornell Chronicle, ACM Showcase.

[C3] Feasibility of Radio Frequency Based Wireless Sensing of Lead Contamination in Soil
Gao, Y., Ahmed, T., Cheng, Z., Mohammed, M. & Nandakumar, R. [open access online]

- In Proc. of the *21st International Conference on Embedded Wireless Systems and Networks*, December 10-13, 2024, Abu Dhabi, UAE. (EWSN'24)
- Acceptance rate: 22.86%.
- News: Best Paper Award, Cornell Chronicle, Phys.org, MSN, American Technion Society, KARMACTIVE.

[J1] Biomimicry in Nanotechnology: a Comprehensive Review
Himel, M. H., Sikder, B., Ahmed, T. & Choudhury, S. M. [open access online]

- In *Nanoscale Advances* 5, no. 3 (2023): 596-614.

[C2] COVID-19 Identification From Lung CT Scans in a Low-Resource Setting Using a Regularized 3D Convolutional Neural Network
Ahmed, T., Nakib, M., Haque, M. A., & Miah, M. M. M. [pre-print] | [IEEE Xplore library]

- In Proc. of the *12th International Conference on Electrical and Computer Engineering*, December 21-23, 2022, Dhaka, Bangladesh. (ICECE)

[C1] Epileptic Seizure Prediction Using Bandpass Filtering and Convolutional Neural Network
Mustaqeem, N., Rahman, T., Priyo, J. F. B. K., Parvez, M. Z., & Ahmed, T. [pre-print] | [Springer library]

- In Proc. of the *1st International Conference on Machine Intelligence and Emerging Technologies*, Spetember 23-25, Noakhali, Bangladesh. (MIET)

ACADEMIC SERVICE

- Reviewer: ACM CHI ['25, '26], ACM/IEEE HRI ['26], ACM DIS ['25, '26], UbiComp/ISWC ['25], IEEE Signal Processing Letters ['25, '26], ACM MobileHCI ['26]
- Topic Chair: IEEE ECCE ['25]
- Student Volunteer: ACM HotMobile ['25]

WORK EXPERIENCE

Cornell Tech NYC, US

- Graduate Research Assistant @ Wireless Sensing and Mobile Systems Lab Aug 2023 - *Present*
- Graduate Teaching Assistant: ECE 5260/ORIE 5735 Graph-Based Data Science for Networked Systems (Spring 25), CS/INFO 5304 Data Science in the Wild (Spring 24), INFO 5600 AI for Healthcare (Fall 23, 24), INFO 5375 Machine Learning for Health (Fall 26).

Environmental Defense Fund NYC, US

- Research Fellow @ MethaneSAT Data Platform May 2026 - Aug 2026
  Manager: Sasha Ayvazov.

Brac University Dhaka, Bangladesh

- Lecturer @ Department of Computer Science and Engineering Jan 2020 - Aug 2023
  Courses Instructed: CSE 460 VLSI Design, CSE 428 Image Processing, CSE 350 Digital Electronics and Pulse Techniques, CSE 251 Electronic Devices and Circuits, CSE 250 Circuits and Electronics.

SCHOOL PROJECTS

- Peer-to-peer computational resource sharing in networked edge devices [report]
- Impact of multi-avatar and camera perspective on self-presence in VR [report]
- Performance analysis of privacy-preserving logistic regression classifiers on the MNIST dataset [report] [code]
- Segmentation of ground-glass opacity from COVID-19-infected lung CT scans [report] [code]
- Sign-language digit classification with explainable AI [report] [code]
- Stage spotlight automation using deep learning and micro-controllers [code]
- Design, implementation & verification of 32-bit MIPS processor [code], 8×8 Booth-encoded multiplier [report]
- Real-time ECG monitoring and disease detection using Arduino, ECG chip, and WiFi module [report]

TECHNICAL SKILLS

Programming Languages:	Python, C, C++, Assembly, Verilog
Machine Learning Libraries:	TensorFlow, Keras, PyTorch, Scikit-learn, MLX
Circuit Design & Simulation:	PSpice, Proteus, Quartus, Cadence
Numerical Analysis:	MATLAB, Numpy, SciPy
Google Workspace:	Docs, Sheets, Slides, Colab
Writing & Presentation:	Microsoft Word, Microsoft PowerPoint, LaTeX
Millimeter wave:	mmWave Studio
USRP:	GNU Radio
Sleep Medicine:	Rem Logic

HONORS & AWARDS

- PiTech Impact Fellowship ($20000), Cornell Tech — 2026
- Digital Life Initiative (DLI) Doctoral Fellowship ($5000), Cornell Tech — 2025-26
- Student Travel Grant ($800), ACM HotMobile '25, California, USA — 2025
- Best Paper Award (€1200), EWSN'24, Abu Dhabi, UAE — 2024
- Student Travel Grant (€2000), EWSN'24, Abu Dhabi, UAE — 2024
- Photo Contest Winner, "Tech Innovation in Frame" category, Cornell Tech, USA — 2023
- Bachelor's Degree with Honors (GPA $\geq$ 3.75), BUET — 2019
- Dean's List (Academic Year GPA $\geq$ 3.75), BUET — 2015, 2016, 2018
- University Merit Award, BUET — 2015, 2016, 2017, 2018
- National Idea Competition, Top 10, Ministry of Power, Energy & Mineral Resources, Bangladesh — 2017
- National Physics Olympiad, 1^{st} position, St. Joseph Higher Secondary School, Dhaka, Bangladesh — 2014
- Bangladesh Physics Olympiad, 7^{th} position, Dhaka Divisional, Bangladesh — 2014